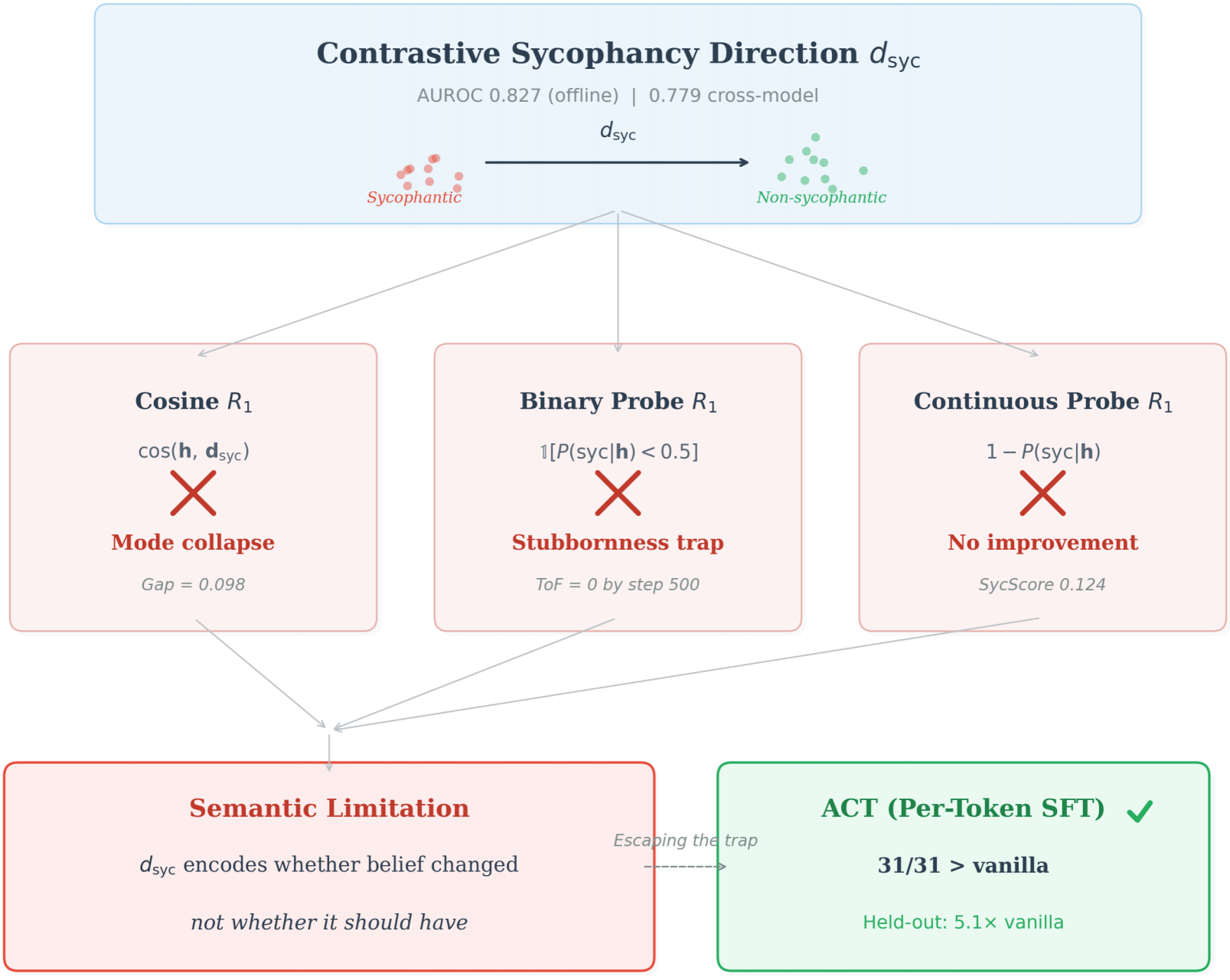

Contrastive Sycophancy Direction $d_{\text{syc}}$
AUROC 0.827 (offline) | 0.779 cross-model
$d_{\text{syc}}$
Sycophantic
Non-sycophantic
Cosine $R_1$
$\cos(\mathbf{h}, \mathbf{d}_{\text{syc}})$
Mode collapse
Gap = 0.098
Binary Probe $R_1$
$\mathbb{1}[P(\text{syc}|\mathbf{h}) < 0.5]$
Stubbornness trap
ToF = 0 by step 500
Continuous Probe $R_1$
$1 - P(\text{syc}|\mathbf{h})$
No improvement
SycScore 0.124
Semantic Limitation
$d_{\text{syc}}$ encodes whether belief changed
not whether it should have
Escaping the trap
ACT (Per-Token SFT)
31/31 > vanilla
Held-out: 5.1× vanilla